Results for algebra2005

Three training-graph variants; six models per variant; N = 5, 10, ..., 50.

Higher is better for Ada, NOV and Ep. Bold = best; underline = second best within each row.

Rankings use unrounded values. All 18 model runs completed for this dataset.

Training-graph variant	Retained exfr edges	Share of source	Mean forgetting
1. Remove exfr0.00 training edges	258,120 / 466,120	55.4%	0.196
2. Weighted nonzero sampling, up to 400/student	146,875 / 466,120	31.5%	0.243
3. Weighted nonzero sampling, up to 200/student	75,932 / 466,120	16.3%	0.267

Main model across the three variants

Variant	Ada@10	Ada@50	NOV@50	Ep0.5@50	Ep0.7@50
Remove exfr0.00 training edges	0.745	0.600	0.986	0.212	-0.024
Weighted nonzero sampling, up to 400/student	0.748	0.594	0.987	0.299	0.216
Weighted nonzero sampling, up to 200/student	0.741	0.595	0.985	0.071	-0.123

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 258,120 / 466,120 (55.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.758	0.707	**0.788**	0.740	0.740	0.707
10	0.745	0.699	**0.766**	0.705	0.716	0.688
15	**0.734**	0.591	0.729	0.668	0.717	0.665
20	0.717	0.550	**0.718**	0.637	0.717	0.646
25	0.687	0.514	**0.703**	0.613	0.687	0.626
30	0.658	0.493	**0.690**	0.596	0.658	0.610
35	0.647	0.496	**0.673**	0.583	0.636	0.597
40	0.630	0.480	**0.639**	0.573	0.616	0.587
45	**0.620**	0.469	0.620	0.564	0.602	0.575
50	0.600	0.453	**0.601**	0.554	0.588	0.564

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.991	0.985	**0.994**	0.992	0.987	0.990
10	0.988	0.984	**0.992**	0.989	0.983	0.991
15	0.988	0.983	0.988	**0.989**	0.983	0.988
20	0.986	0.982	0.987	0.988	0.985	**0.988**
25	0.987	0.981	0.988	0.986	0.987	**0.988**
30	0.987	0.980	0.986	0.986	0.988	**0.988**
35	0.987	0.980	0.988	0.986	**0.988**	0.987
40	0.987	0.980	**0.988**	0.987	0.987	0.986
45	0.987	0.980	**0.987**	0.986	0.986	0.987
50	0.986	0.980	**0.988**	0.986	0.985	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 258,120 / 466,120 (55.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.009	**0.052**	-0.104	-0.044	-0.086	0.011
10	0.034	**0.108**	-0.110	-0.039	0.055	0.058
15	0.108	0.239	**0.285**	0.040	0.102	0.126
20	0.092	0.119	**0.160**	0.121	0.034	0.151
25	0.214	0.038	**0.252**	0.162	0.136	0.174
30	0.062	-0.079	**0.317**	0.196	0.018	0.203
35	0.186	0.064	**0.311**	0.220	0.208	0.234
40	0.216	0.112	**0.441**	0.224	0.198	0.207
45	**0.273**	0.144	-0.010	0.233	0.149	0.216
50	0.215	0.000	-0.129	**0.220**	0.161	0.188

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.054	**0.014**	-0.110	-0.056	-0.107	-0.011
10	0.027	**0.086**	-0.204	-0.073	0.051	0.025
15	0.108	0.235	**0.275**	-0.024	0.102	0.100
20	0.079	0.117	**0.159**	0.056	0.025	0.125
25	0.205	0.025	**0.252**	0.122	0.034	0.157
30	0.043	-0.180	**0.317**	0.175	-0.092	0.185
35	0.133	0.027	**0.311**	0.202	0.166	0.220
40	0.199	0.101	**0.441**	0.209	0.183	0.199
45	**0.265**	0.140	-0.107	0.219	0.138	0.207
50	**0.212**	-0.014	-0.156	0.208	0.155	0.175

Group 1: Remove exfr0.00 training edges

Retained exfr edges: 258,120 / 466,120 (55.4%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.217	-0.104	-0.313	-0.136	-0.218	**-0.090**
10	-0.081	-0.030	-0.435	-0.185	**-0.012**	-0.080
15	0.016	**0.198**	-0.060	-0.161	0.014	-0.015
20	-0.099	0.102	**0.103**	-0.121	-0.048	0.019
25	-0.013	-0.003	**0.205**	-0.068	-0.035	0.055
30	-0.099	-0.223	**0.185**	0.012	-0.198	0.104
35	0.033	-0.049	**0.196**	0.083	-0.001	0.151
40	0.124	0.051	**0.416**	0.123	0.130	0.145
45	**0.223**	0.124	-0.162	0.149	0.083	0.143
50	**0.195**	-0.030	-0.349	0.151	0.122	0.122

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.305	**-0.201**	-0.409	-0.221	-0.347	-0.205
10	-0.324	**-0.180**	-0.467	-0.305	-0.184	-0.233
15	-0.291	**0.066**	-0.089	-0.279	-0.236	-0.188
20	-0.283	**-0.008**	-0.042	-0.231	-0.368	-0.171
25	-0.185	**-0.108**	-0.108	-0.205	-0.294	-0.146
30	-0.267	-0.324	**0.067**	-0.170	-0.350	-0.114
35	-0.236	-0.195	**0.157**	-0.122	-0.191	-0.062
40	-0.186	-0.085	**0.371**	-0.107	-0.095	-0.070
45	-0.023	**0.023**	-0.382	-0.100	-0.146	-0.055
50	**-0.024**	-0.118	-0.497	-0.082	-0.098	-0.091

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 146,875 / 466,120 (31.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.762	0.703	**0.788**	0.745	0.742	0.715
10	0.748	0.666	**0.766**	0.714	0.735	0.703
15	0.743	0.610	**0.744**	0.686	0.725	0.690
20	**0.731**	0.571	0.718	0.669	0.714	0.679
25	0.694	0.542	**0.703**	0.647	0.692	0.660
30	0.667	0.511	**0.702**	0.627	0.661	0.642
35	0.643	0.485	**0.673**	0.608	0.640	0.626
40	0.623	0.464	**0.648**	0.592	0.624	0.610
45	0.611	0.448	**0.616**	0.578	0.608	0.596
50	0.594	0.439	**0.601**	0.567	0.591	0.580

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.991	0.986	**0.994**	0.989	0.988	0.991
10	0.989	0.985	**0.992**	0.986	0.987	0.990
15	0.989	0.985	**0.990**	0.986	0.986	0.989
20	0.986	0.983	0.987	0.986	0.985	**0.988**
25	0.985	0.982	0.988	0.986	0.985	**0.988**
30	0.987	0.980	0.987	0.986	0.986	**0.988**
35	0.987	0.980	0.988	0.986	0.986	**0.988**
40	0.987	0.978	0.987	0.986	0.987	**0.988**
45	0.987	0.978	**0.987**	0.985	0.987	0.987
50	0.987	0.978	**0.988**	0.985	0.987	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 146,875 / 466,120 (31.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.109	**0.032**	-0.104	-0.088	-0.079	-0.013
10	0.044	**0.060**	-0.079	-0.000	0.032	0.045
15	-0.007	0.003	0.121	0.063	**0.122**	0.085
20	0.062	0.101	**0.336**	0.138	-0.071	0.130
25	0.236	0.150	**0.325**	0.203	0.220	0.162
30	**0.273**	0.097	0.163	0.225	0.247	0.185
35	0.198	0.110	**0.329**	0.248	0.206	0.202
40	**0.269**	0.101	0.140	0.267	0.096	0.220
45	**0.280**	0.087	0.081	0.273	0.224	0.210
50	**0.300**	0.089	0.166	0.261	0.272	0.197

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.158	**-0.007**	-0.110	-0.110	-0.110	-0.048
10	0.018	**0.029**	-0.132	-0.062	0.004	0.004
15	-0.008	-0.014	0.108	0.008	**0.118**	0.068
20	0.062	0.086	**0.336**	0.101	-0.074	0.111
25	0.233	0.143	**0.325**	0.174	0.218	0.142
30	**0.273**	0.081	0.163	0.208	0.244	0.171
35	0.198	0.102	**0.329**	0.232	0.204	0.191
40	**0.255**	0.090	0.140	0.252	0.082	0.207
45	**0.275**	0.073	0.081	0.265	0.195	0.197
50	**0.299**	0.076	0.166	0.255	0.256	0.189

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 2: Weighted nonzero sampling, up to 400/student

Retained exfr edges: 146,875 / 466,120 (31.5%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.305	**-0.142**	-0.313	-0.221	-0.234	-0.155
10	-0.199	**-0.129**	-0.408	-0.215	-0.178	-0.146
15	-0.221	-0.116	-0.285	-0.134	**0.060**	-0.092
20	-0.142	-0.014	**0.305**	-0.056	-0.132	-0.027
25	0.090	0.110	**0.203**	0.033	0.103	0.024
30	**0.190**	0.052	0.047	0.102	0.124	0.082
35	-0.013	0.073	**0.202**	0.149	0.146	0.128
40	**0.215**	0.061	-0.031	0.188	0.006	0.156
45	**0.258**	0.040	-0.000	0.213	0.141	0.151
50	**0.288**	0.040	0.143	0.205	0.209	0.132

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.392	**-0.225**	-0.409	-0.293	-0.348	-0.255
10	-0.360	**-0.261**	-0.445	-0.305	-0.327	-0.283
15	-0.388	-0.294	-0.312	-0.255	**-0.138**	-0.265
20	-0.319	-0.182	**0.067**	-0.204	-0.363	-0.221
25	**0.011**	-0.045	-0.036	-0.147	-0.180	-0.191
30	**0.053**	-0.035	-0.174	-0.093	-0.099	-0.140
35	-0.119	-0.021	**0.034**	-0.053	-0.111	-0.096
40	**0.007**	-0.019	-0.107	-0.006	-0.245	-0.051
45	**0.205**	-0.033	-0.298	0.026	-0.152	-0.056
50	**0.216**	-0.078	-0.280	0.033	0.001	-0.070

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 75,932 / 466,120 (16.3%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(a) Adaptability (Ada)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.748	0.711	**0.788**	0.701	0.768	0.712
10	0.741	0.702	**0.766**	0.686	0.748	0.707
15	0.730	0.648	**0.740**	0.666	0.736	0.690
20	0.723	0.620	0.718	0.648	**0.730**	0.672
25	0.688	0.602	0.703	0.634	**0.709**	0.653
30	0.680	0.572	**0.702**	0.622	0.684	0.633
35	0.666	0.538	**0.673**	0.611	0.654	0.615
40	0.644	0.506	**0.644**	0.599	0.634	0.601
45	0.614	0.480	**0.616**	0.586	0.606	0.586
50	0.595	0.458	**0.601**	0.572	0.595	0.572

(b) Novelty (NOV)

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	0.989	0.986	**0.994**	0.993	0.990	0.991
10	0.986	0.984	**0.992**	0.988	0.987	0.988
15	0.985	0.983	**0.989**	0.986	0.986	0.986
20	0.986	0.984	**0.987**	0.987	0.986	0.986
25	0.986	0.983	**0.988**	0.986	0.987	0.986
30	0.987	0.981	**0.987**	0.987	0.986	0.986
35	0.987	0.980	**0.988**	0.987	0.986	0.987
40	0.986	0.978	**0.988**	0.986	0.986	0.986
45	0.985	0.977	**0.987**	0.985	0.987	0.986
50	0.985	0.977	**0.988**	0.985	0.987	0.986

Seed 2024; single-run values. No across-seed standard deviation is available.

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 75,932 / 466,120 (16.3%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(c) Ep, response threshold = 0.4

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.079**	0.041	-0.104	0.025	-0.114	-0.013
10	-0.066	**0.081**	-0.070	0.028	-0.038	0.055
15	0.000	**0.174**	0.172	0.092	0.058	0.107
20	0.099	0.109	**0.276**	0.186	0.148	0.150
25	**0.323**	0.104	0.302	0.195	0.307	0.189
30	0.231	0.196	**0.247**	0.210	0.174	0.218
35	0.224	0.046	**0.363**	0.221	0.308	0.218
40	0.070	0.018	0.228	0.203	**0.279**	0.224
45	**0.353**	0.056	0.037	0.212	0.351	0.186
50	0.073	0.021	0.058	0.186	**0.291**	0.202

(d) Ep, response threshold = 0.5

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	**0.043**	0.001	-0.110	0.008	-0.219	-0.041
10	-0.153	**0.053**	-0.166	-0.010	-0.110	0.026
15	-0.021	**0.170**	0.161	0.044	-0.068	0.076
20	0.033	0.107	**0.275**	0.147	0.136	0.126
25	**0.306**	0.103	0.302	0.172	0.304	0.175
30	0.223	0.195	**0.247**	0.196	0.173	0.201
35	0.219	0.029	**0.363**	0.207	0.308	0.209
40	0.054	-0.039	0.228	0.192	**0.279**	0.216
45	**0.350**	0.007	0.037	0.202	0.345	0.175
50	0.071	-0.023	0.058	0.173	**0.286**	0.193

Group 3: Weighted nonzero sampling, up to 200/student

Retained exfr edges: 75,932 / 466,120 (16.3%) | theta_m=0.8, theta_f=0.8, rec=10 | 6/6 models completed

(e) Ep, response threshold = 0.6

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.140	-0.117	-0.313	**-0.055**	-0.340	-0.139
10	-0.264	**-0.092**	-0.414	-0.105	-0.206	-0.099
15	-0.201	**0.125**	-0.035	-0.082	-0.255	-0.063
20	-0.156	0.089	**0.245**	0.023	0.019	0.002
25	**0.262**	0.089	0.211	0.067	0.205	0.069
30	0.201	0.187	**0.230**	0.103	0.148	0.125
35	0.113	-0.007	0.157	0.131	**0.192**	0.149
40	-0.014	-0.068	0.188	0.104	**0.208**	0.161
45	**0.337**	-0.033	-0.073	0.118	0.328	0.122
50	0.063	-0.105	-0.049	0.100	**0.275**	0.152

(f) Ep, response threshold = 0.7

N	2CKG4ER	No-mastery	No-forgetting	No-student-feature	No-exercise-feature	ID-based only
5	-0.245	-0.209	-0.409	**-0.142**	-0.402	-0.245
10	-0.352	-0.234	-0.446	-0.239	-0.414	**-0.231**
15	-0.357	**-0.034**	-0.177	-0.212	-0.350	-0.225
20	-0.359	-0.130	**0.074**	-0.144	-0.252	-0.193
25	**0.084**	-0.105	-0.051	-0.123	-0.057	-0.125
30	-0.099	**0.092**	0.021	-0.115	-0.049	-0.083
35	-0.133	-0.072	**0.037**	-0.081	-0.017	-0.070
40	-0.321	-0.170	-0.044	-0.103	-0.106	**-0.040**
45	**0.233**	-0.136	-0.396	-0.084	0.115	-0.070
50	-0.123	-0.196	-0.372	-0.110	**0.050**	-0.042